FIITJEE

ALL INDIA TEST SERIES

FULL TEST – XI

JEE (Main)-2024

TEST DATE: 31-03-2024

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts.**
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A and Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B :** Do any five questions out of 10 questions.

Section-A (01 – 20, 31 – 50, 61 – 80) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Section-B (21 – 30, 51 – 60, 81 – 90) *contains 30 Numerical based questions. The answer to each question is rounded off to the nearest integer value. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

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Physics

PART – A

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. Current density in a cylindrical wire of radius R is given as

$$J=\begin{cases} J_0\left(\frac{x}{R}-1\right) & \text{for} \quad 0 \le x \le R/2 \\ J_0\left(\frac{x}{R}\right) & \text{for} \quad \frac{R}{2} \le x \le R \end{cases}$$

The current in wire is

(A) $\frac{7}{24}\pi J_0 R^2$ (B) $\frac{1}{6}\pi J_0 R^2$

(C) $\frac{7}{12}\pi J_0 R^2$ (D) $\frac{5}{12}\pi J_0 R^2$

2. A particle of mass m moves in a circular orbit in a central potential field $U(r)=\frac{k}{2}r^2$. If Bohr's quantisations are applied, radii of possible orbitals and energy levels vary with quantum number n as

(A) $r_n \propto n,\ E_n \propto n$ (B) $r_n \propto n,\ E_n \propto \frac{1}{n}$

(C) $r_n \propto \sqrt{n},\ E_n \propto n$ (D) $r_n \propto n^2,\ E_n \propto \frac{1}{n^2}$

3. One mole of oxygen is contained in a rigid container with walls of inner surface area A, thickness d and thermal conductivity k. The gas is at initial temperature T_0 and the surrounding temperature of air is $2T_0$. The temperature of the gas at time t is (R is gas constant)

(A) $T=T_0\left[2-e^{-\frac{2kAt}{5dR}}\right]$ (B) $T=T_0\left[1-e^{-\frac{2kAt}{5dR}}\right]$

(C) $T=T_0 e^{-\frac{2kAt}{5dR}}$ (D) $T=T_0 e^{-\frac{kAt}{dR}}$

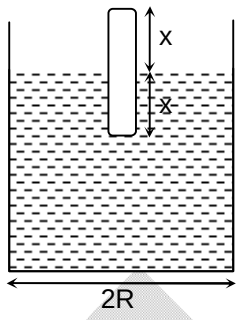
4. The force acting on a particle of mass m is given by $F=mA\frac{e^{-2r}}{r^3}$. The maximum percent error in the measurement of acceleration of the particle at $r=(1\pm 0.05)\mu m$ is (A is a constant in proper unit)

(A) 5% (B) 15%

(C) 25% (D) 35%

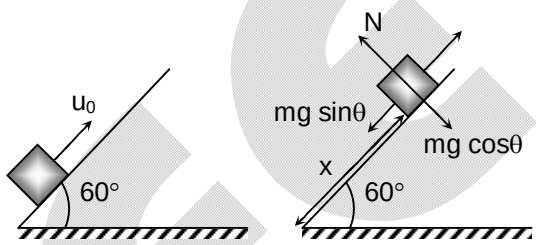
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5. A candle of cross-sectional radius r floats in water in a cylindrical container of radius (R >> r) as shown in the figure. Half the length of the candle always remains submerged in water. If it is burning at the rate of 4 mmh^{-1}, then the top of the candle will



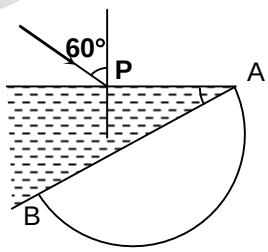
(A) remain at the same height
(B) fall at the rate of $2mmh^{-1}$
(C) fall at the rate of $4mmh^{-1}$
(D) go up at the rate of $2mmh^{-1}$

6. A worker is throwing a box up a rough plank inclined at 60° above the horizontal, so that the box leaves the bottom of the plank at speed u_0. The coefficient of friction (assuming kinetic and static friction are almost equal) increases with the distance along the plank such that $\mu = kx$, where k = 15 m/s. Now, once the box comes to rest, it will remain at rest only if the least initial speed is ($g = 10\ m/s^2$)



(A) $\sqrt{5}$ m/s
(B) $\sqrt{2}$ m/s
(C) $\sqrt{3}$ m/s
(D) $\sqrt{7}$ m/s

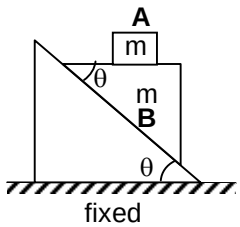
7. A hemisphere of radius R and refractive index $\sqrt{2}$ is placed in a liquid of refractive index $\sqrt{3}$ as shown in the figure. A ray is incident at point P at an angle 60°. As the ray emerges out from the hemisphere the angle of refraction is 0°. The distance of the point P from A is



(A) $\sqrt{2}R$
(B) $\frac{2R}{\sqrt{3}}$
(C) $\sqrt{3}R$
(D) 2R

8. In a screw gauge, there are 100 divisions in circular scale and on one complete rotation of circular scale, the main scale moves by 1 division. When there is no gap between the jaws, 97th division coincides with the main scale zero and zero of main scale is not visible. While measuring the diameter of a ball, the circular scale is between 3 mm mark and 4 mm mark such that the 76th division of circular scale coincides with the reference line. Select the correct alternative.
(A) The least count of the screw gauge is 0.01 cm.
(B) The zero error is –0.04 mm
(C) The diameter of the ball is 3.79 mm
(D) The main scale reading is 4 mm

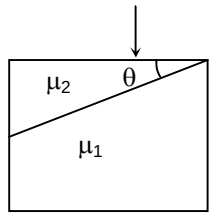
9. Consider the arrangement shown in the figure. All surfaces are smooth. At certain instant the velocity of the wedge B is v m/s and acceleration is 'a' then the power of normal force on the block A at that instant is :



(A) $-mvg\sin^2\theta$
(B) $-mv(g - a\sin\theta)\sin\theta$
(C) $-mvg\cos^2\theta$
(D) $-mv(g - a)\sin^2\theta$

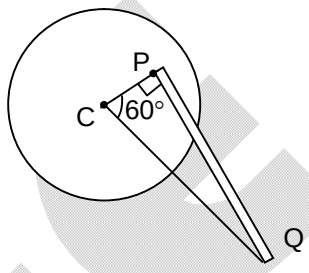
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10. A ray of light falls perpendicularly at the surface of composite glass slab, made of two transparent prism of small angle of refractive index μ_1 and μ_2 respectively as shown in the figure. The correct deviation S is



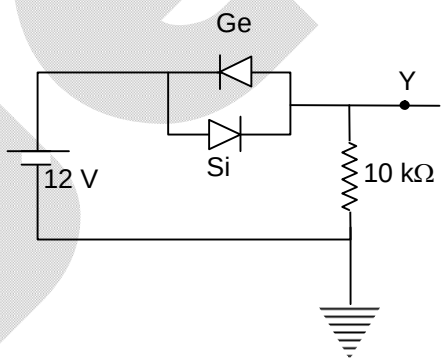
(A) zero
(B) $(\mu_1 + \mu_2 - 2)\theta$
(C) $(\mu_2 - \mu_1)\theta$
(D) $\left(1-\frac{\mu_1}{\mu_2}\right)\theta$

11. A magnetic field, confined in a cylindrical region of radius R, is changing at the rate of 4 T/s. A conducting rod PQ of length $\sqrt{\frac{3}{2}}R$ is placed in the region as shown in the figure. The induced emf across the rod will be



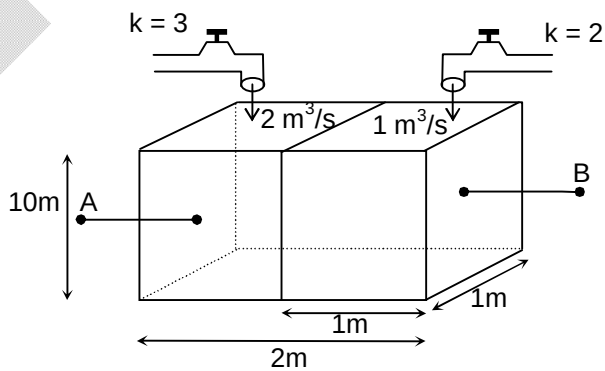
(A) $\frac{R^2}{24}(\pi+6)$ Volts
(B) $\frac{R^2\pi}{24}$ Volts
(C) $\frac{R^2\pi}{24}$ Volts
(D) $\frac{R^2}{6}(\pi+6)$ Volts

12. Two junction diode one of germanium (Ge) and other of Silicon (Si) are connected as shown in the figure to a battery of emf 12V and a load resistance 10 kΩ. The germanium diode conducts at 0.3 V and silicon diode at 0.7 V. When a current flows in the circuit, the potential of terminal Y will be



(A) 12V
(B) 11V
(C) 11.3V
(D) 11.7 V

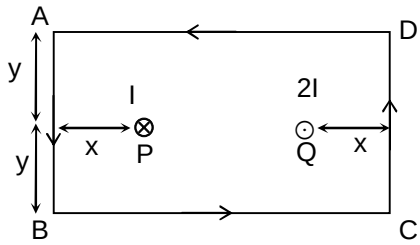
13. A thin metallic partition of negligible thickness is inserted between two shaded metallic plates as shown in the figure. The remaining ends are then packed with insulating plates to form a container like structure. Two taps shown are opened at t = 0 and finally closed at t = 5 sec. Find capacitance of system between A and B after closing taps. (Assuming liquid to be non-conducting) volumetric flow rates and dielectric constants of liquid are given.



(A) 8.85×10^{-11} F
(B) 8.85×10^{-10} F
(C) 4.42×10^{-11} F
(D) 4.42×10^{-10} F

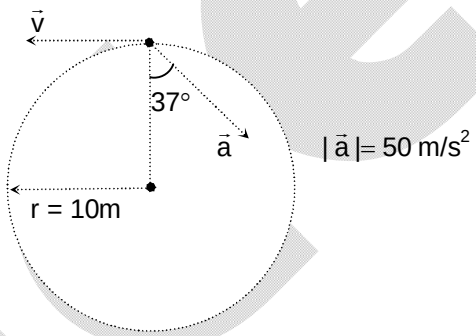
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14. Let B_P and B_Q be the magnetic field produced by the wire P and Q which are placed symmetrically in a rectangular loop ABCD as shown in the figure. Current in wire P is I directed inwards and in Q is 2I directed outwards. If $\int_A^B \vec{B}_Q \cdot \vec{d\ell} = 2\mu_0$, $\int_D^A \vec{B}_P \cdot \vec{d\ell} = -2\mu_0$ and $\int_A^B \vec{B}_P \cdot \vec{d\ell} = -\mu_0$ Tesla-meter, the value of I will be



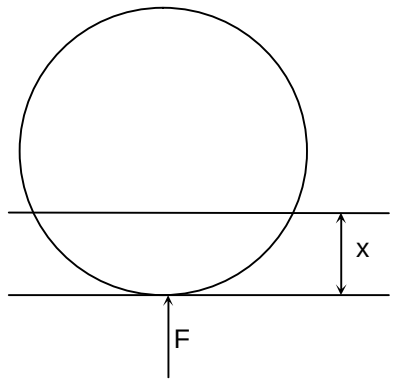
(A) 8A (B) 4A
(C) 5A (D) 6A

15. In the figure the particle is travelling counterclockwise in a circle of radius 10 m. The acceleration vector is indicated at a specific time. Find the value of 'v' at this time.



(A) 10 m/s
(B) 15 m/s
(C) 20 m/s
(D) 7 m/s

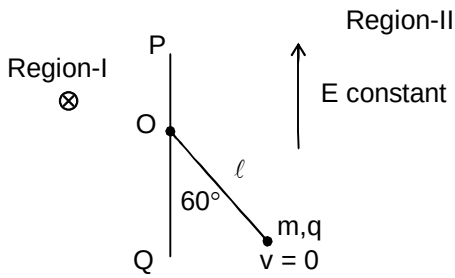
16. Experimentally it has been found that the force F needed to compress an elastic ball, elastically through a distance x (as shown in the figure) follows the formula $F = ax + bx^2 + cx^3$ where a, b, c and constants. The small ball of mass m at rest is dropped from a great height h. It bounces elastically off the floor and is compressed to a maximum distance 'd' during the bounce. The height h is (g is the acceleration due to gravity)



(A) $\frac{1}{mg}\left[\frac{1}{2}ad^2 + \frac{1}{3}bd^3 + \frac{1}{4}cd^4\right]$ (B) $\frac{1}{mg}\left[ad + bd^2 + cd^3\right]$

(C) $\frac{1}{mg}\left[ad^2 + bd^3 + cd^4\right]$ (D) $\frac{1}{mg}\left[ad^2 + 2bd^2 + 3cd^4\right]$

17. Consider a bob of mass m and having charge q attached with a light string of length ℓ and pivoted at point O. It is released at rest at 60° with vertical. There are two regions. Region-I (left of line PQ) has a uniform and constant magnetic field B directed inside plane of paper. Region-II (right of line PQ) has a constant and uniform electric field E directed vertically up as shown in the figure. Consider no effect of gravity in both the regions. Which of the following is correct?



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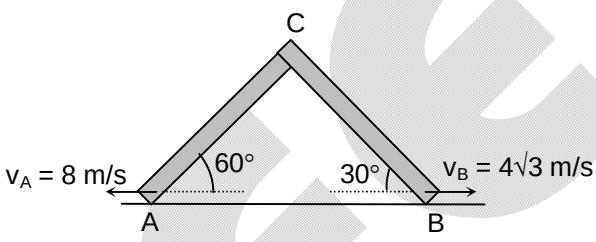
(A) Time taken by a particle to cross region-I for first time is $\pi\sqrt{\frac{2m\ell}{5qE}}$

(B) Time taken by a particle to cross region-I for first time is $\pi\sqrt{\frac{2m\ell}{13qE}}$

(C) Time taken by a particle to cross region-I for first time is $\sqrt{\frac{5qE}{2m\ell}}$

(D) Time taken by a particle to cross region-I for first time is $\sqrt{\frac{13qE}{m\ell}}$

18. Two rods AC and BC are hinged at the end C and the ends A and B are moving along the surface with velocity $v_A = 8$ m/s and $v_B = 4\sqrt{3}$ m/s, respectively as shown in the figure. The velocity of hinged point 'C' is:



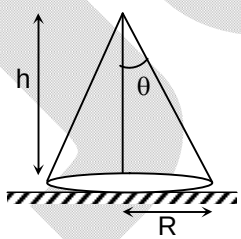
(A) $\sqrt{13}$ m/s

(B) $3\sqrt{13}$ m/s

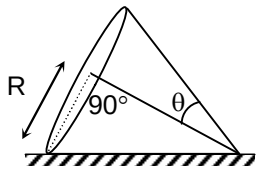
(C) $2\sqrt{13}$ m/s

(D) $4\sqrt{13}$ m/s

19. A solid cone is placed on a horizontal surface with height 'h' and radius 'R'. Its apex angle is θ. To change the position of the cone as shown in the figure-A to figure-B, no work is required, then



figure(A)



figure(B)

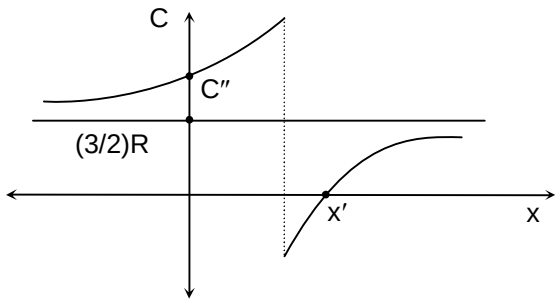
(A) $\theta = \sin^{-1}\left(\frac{1}{3}\right)$

(B) $\theta = \frac{\pi}{4}$

(C) $\theta = \frac{\pi}{3}$

(D) $\theta = \sin^{-1}\left(\frac{1}{5}\right)$

20. One mole of an ideal monoatomic gas is taken along the process in which PV^x = constant. The graph shown represents the variation of molar heat capacity of such a gas with respect to x, The value of C″ and x′ respectively are given by



(A) $\frac{5R}{2}, \frac{5}{2}$

(B) $\frac{7R}{2}, \frac{7}{2}$

(C) $\frac{5R}{2}, \frac{5}{3}$

(D) $\frac{7R}{2}, \frac{7}{5}$

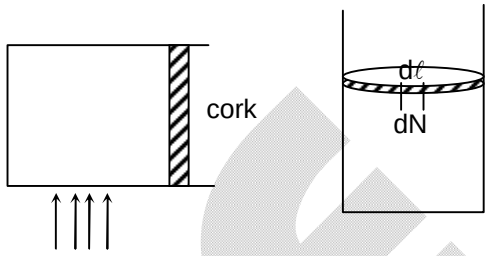
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SECTION – B

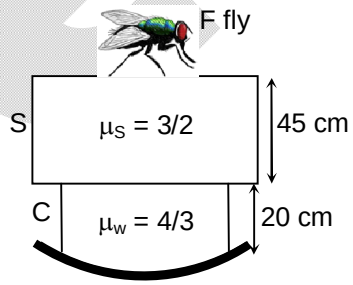
(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

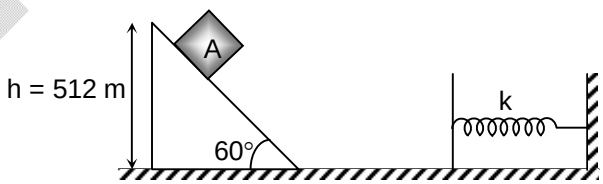
21. Figure shows a cylindrical tube of radius 5 cm and length 20 cm. It is closed by tight fitting cork. The friction coefficient between the cork and the tube is 0.20. The tube contains an ideal gas at a pressure of 1 atm and temperature 300 K. The tube is slowly heated and it is found that the cork pops out when the temperature reaches 600 K. Let denote the magnitude of normal contact force exerted by a small length 'dℓ' of the cork along the periphery. Assuming that the temperature of the gas is uniform at any instant. The value of $\frac{dN}{d\ell} = 1.25 \times 10^{\alpha}$ N/m. Find the value of α.



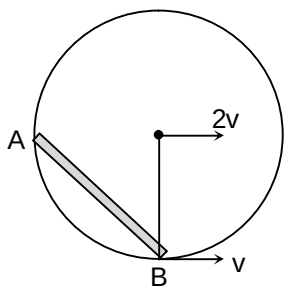
22. A fly F is sitting on a glass slab S, 45 cm thick and of refractive index $\frac{3}{2}$. The slab covers the top of a container C containing water of refractive index $\frac{4}{3}$ upto a height 20 cm. Bottom of the container is closed by a concave mirror M of radius curvature 40 cm. If the final image formed by all refraction and refractions assuming paraxial rays is formed $\frac{45}{x}$ cm below the upper surface of the glass slab then find x.



23. A block A is released form the top of a fixed wedge of height 512 m. All the surfaces are frictionless. A collides with the spring elastically while moving from inclined to plane surface or vice-versa and A never looses contact with surfac.e Collision between block and horizontal surface incline plane are perfectly inelastic. Find the height (in meter) attained by block when it climbs the wedge second time.



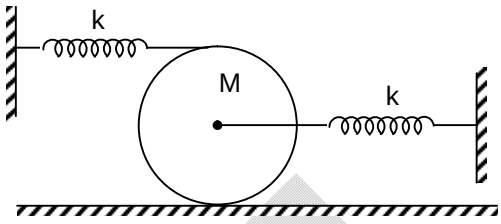
24. A uniform rod AB of mass m and length $\sqrt{2}R$ is moving in a vertical plane inside a hollow sphere of radius R. The sphere is rolling on a fixed horizontal surface without slipping with velocity of its centre of mass 2v. When the end B is at the lowest position, its speed is found to be v as shown in the fiugre. If the kinetic energy of the rod at this instant is $\frac{4}{k}mv^2$. Find the value of k.



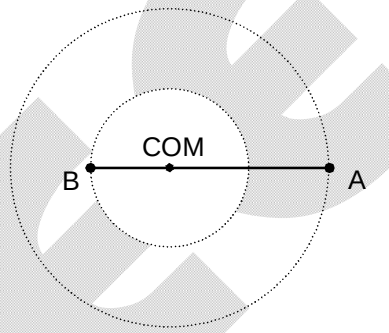
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25. The shortest wavelength of the brackett series of a hydrogen like atom (atomic number = z) is the same as the shortest wavelength of the Balmer series of hydrogen atom. Find the value of z.

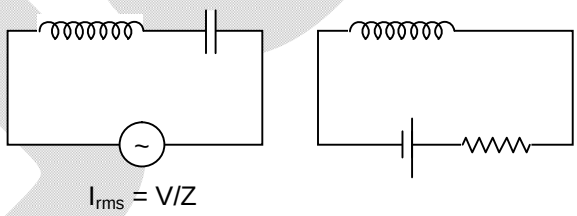
26. A solid uniform cylinder of mass M performs small oscillation due to the action of two springs, each having stiffness k. If time period of these oscillations in absence of any sliding is $T = 2\pi\sqrt{\frac{\alpha M}{10k}}$. Then find the value of α. (spring have their natural length initially)



27. The figure shows a binary star system revolving about their centre of mass. The masses of star A and star B are 15×10^{30} kg and 45×10^{30} kg respectively. Find the ratio of area swept by star A to area swept by star B in a common time interval.



28. A coil, a capacitor and an AC source of voltage 24V (rms) are connected in series. By varying the frequency of the source, a maximum rms current of 6A is observed. If this coil is connected to a DC battery of emf 12V and internal resistance 2Ω, then find the current (in amp) through it in steady state.



29. A person sound of 70 dB at 4m distance away from him. For hearing sound clearly if thereshold sound level is 10 dB, then upto how much distance from person his sound can be heard clearly in kilometer.

30. A sonometer wire resonates with a given tuning fork forming standing waves with five antinodes between the two bridges when a mass of 9 kg is suspended from the wires. When this mass is replaced by mass M, the wire resonates with the same tuning fork forming three antinode for the same position of the bridges. Then find the value of $\sqrt{M}$

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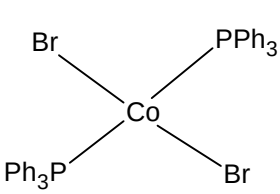
Chemistry

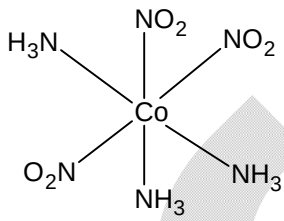
PART – B

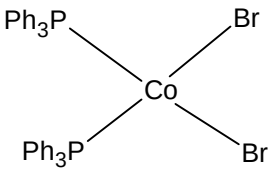
SECTION – A
(One Options Correct Type)

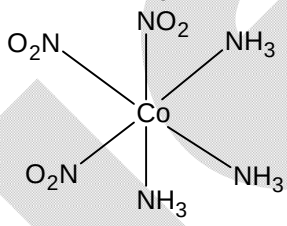
This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

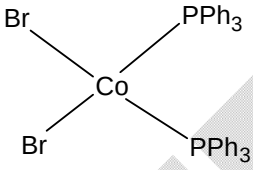
31. The correct structure of trans $\left[CoBr_2(PPh_3)_2\right]$ and meridonial $\left[Co(NH_3)_3(NO_2)_3\right]$ respectively are

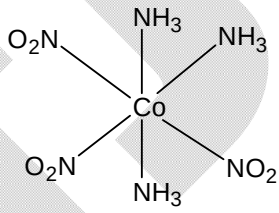
(A) 

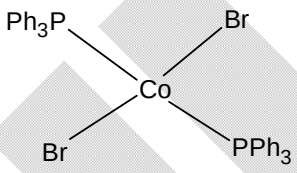


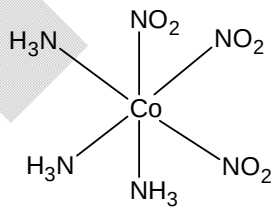
(B) 



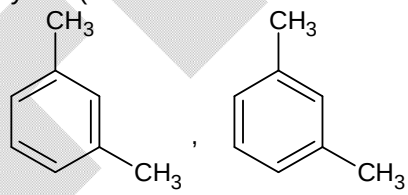
(C) 

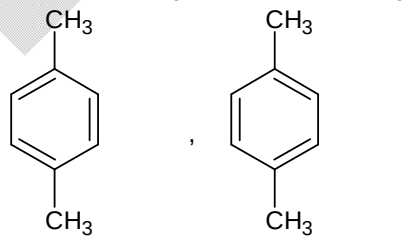


(D) 

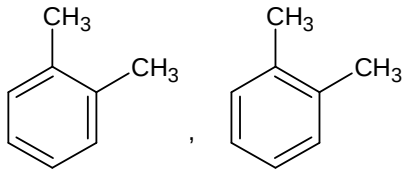


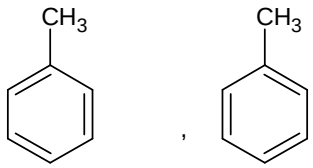
32. Which among the following pairs of structures will give different product after reductive ozonolysis? (Consider the double bond in the structure are rigid and not delocalized).

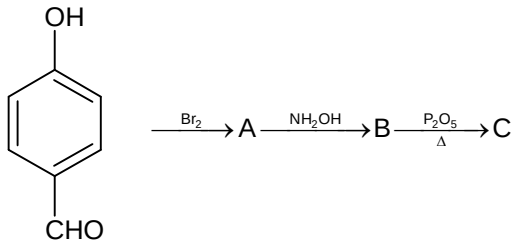
(A) 

(B) 

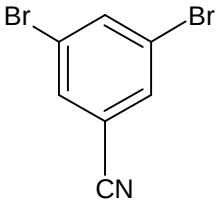
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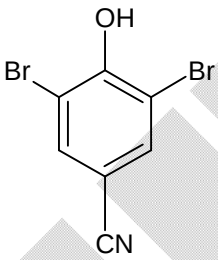
(C) 

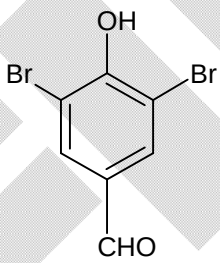
(D) 

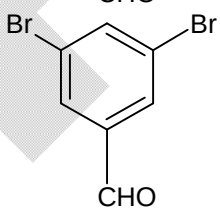
33. 

Consider the above reaction sequence, the product (C) in the sequence is

(A) 

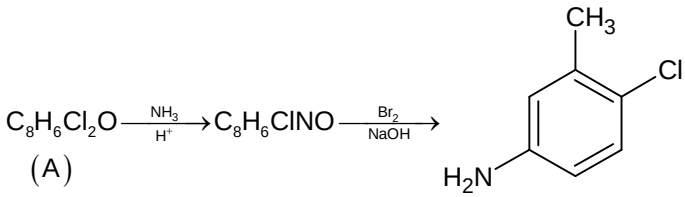
(B) 

(C) 

(D) 

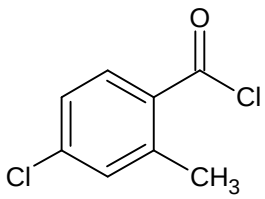
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34.

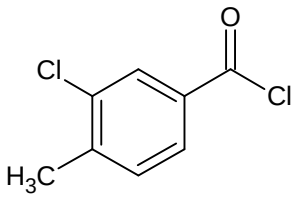


Consider the above reaction, the compound 'A' is:

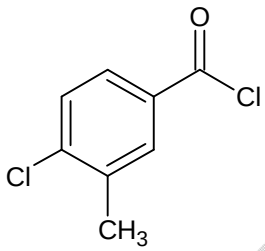
(A)



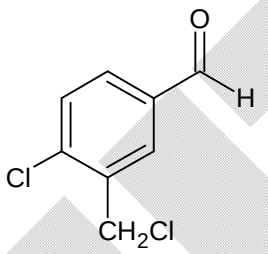
(B)



(C)



(D)



35. In the following pairs, the one in which both transition metal ions are colourless is

(A) Sc^{+3}, Zn^{+2} (B) Ti^{+4}, Cu^{+2}

(C) V^{+2}, Ti^{+3} (D) Zn^{+2}, Mn^{+2}

36. Number of lone pair of electron in the central atom of SCl_2, O_3, ClF_3 and SF_6 respectively, are

(A) 0, 1, 2, 2 (B) 2, 1, 2, 0

(C) 1, 2, 2, 0 (D) 2, 1, 0, 2

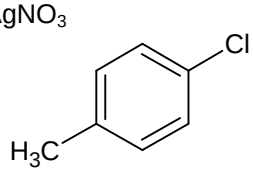
37. Among the following which one forms a yellow precipitate on reacting sequentially with

(1) NaOH

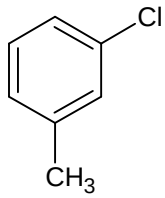
(2) dil. HNO_3

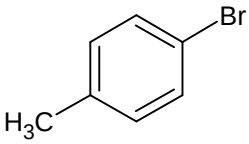
(3) $AgNO_3$

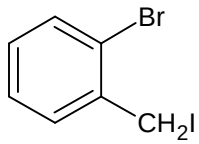
(A)



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(B) 

(C) 

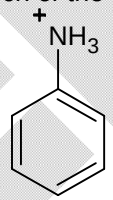
(D) 

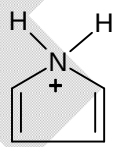
38. The first ionization enthalpy of Na, Mg and Si are 496, 737, 786 kJ mol^{-1} respectively. The first ionization energy (kJ mol^{-1}) of Al is:
(A) 487 (B) 760
(C) 577 (D) 856

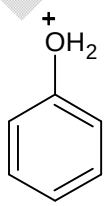
39. Which one of the following complex is violet in colour?
(A) $\left[Fe(SCN)_6\right]^{4-}$ (B) $\left[Fe(CN)_6\right]^{4-}$
(C) $Fe_4\left[Fe(CN)_6\right]_3.H_2O$ (D) $\left[Fe(CN)_5NOS\right]^{4-}$

40. Which of the following pair of molecule contain odd electron molecule and an expanded octet molecule respectively?
(A) BCl_3, SF_6 (B) NO, H_2SO_4
(C) SF_6, H_2SO_4 (D) BCl_3, NO

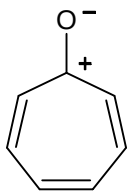
41. A certain orbital has no angular node but two radial nodes. The orbital is
(A) 2s (B) 2p
(C) 3s (D) 3p

42. In which of the following delocalisation of positive charge is possible?
(A) 

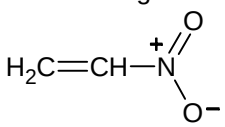
(B) 

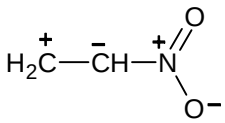
(C) 

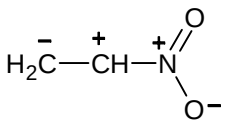
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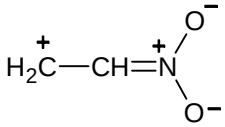
(D) 

43. Least contributing resonance structure of nitroethene is

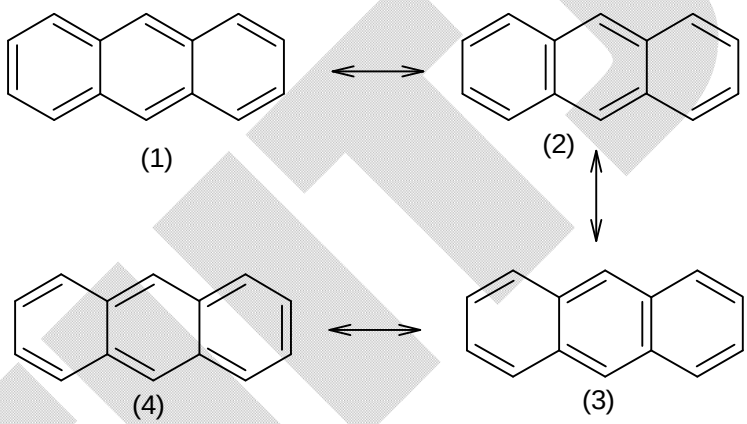
(A) 

(B) 

(C) 

(D) 

44. The correct stability order of given resonating structure is



(A) 1 > 2 > 3 > 4
(B) 4 > 3 > 2 > 1
(C) 1 = 2 = 3 = 4
(D) 2 = 3 > 1 = 4

45. $CH_3 - \overset{+}{C}H - CH = CH_3$ does not involve

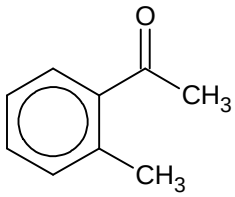
(A) $\sigma - p$ overlap
(B) $\sigma - \pi^*$ overlap
(C) $p\pi - p\pi$ overlap
(D) $p\pi - d\pi$ overlap

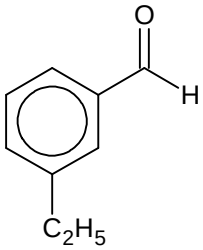
46. For negative electron gain enthalpies of the elements denoted as ΔH_{eg}, the **incorrect** option is

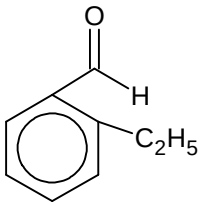
(A) $\Delta H_{eg}(Cl) < \Delta H_{eg}F$
(B) $\Delta H_{eg}(Se) < \Delta H_{eg}(S)$
(C) $\Delta H_{eg}(I) < \Delta H_{eg}(At)$
(D) $\Delta H_{eg}(Te) < \Delta H_{eg}(Po)$

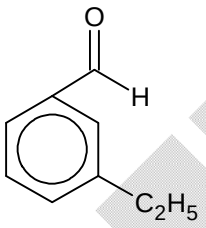
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47. An aromatic compound X with molecular formula $C_9H_{10}O$ gives the following chemical test
(1) forms 2,4-DNP derivative
(2) reduce Tollen's reagent
(3) undergoes Cannizzaro reaction
(4) on vigorous oxidation Phthalic acid is obtained
X is

(A) 

(B) 

(C) 

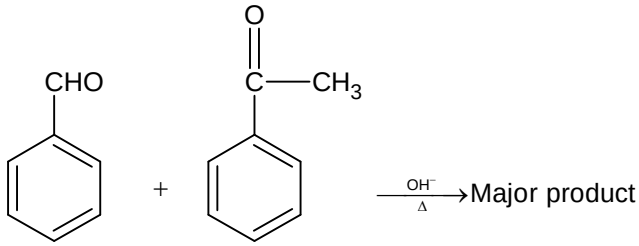
(D) 

48. Which of the following sets of quantum number is not possible?
(A) $n = 2, \ell = 1, m = -1, s = -\frac{1}{2}$
(B) $n = 2, \ell = 0, m = 0, s = -\frac{1}{2}$
(C) $n = 3, \ell = 2, m = -2, s = +\frac{1}{2}$
(D) $n = 3, \ell = 2, m = -3, s = +\frac{1}{2}$

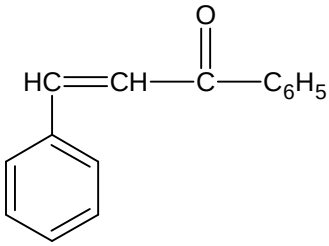
49. Which of the following parameter cannot be used in Born-Haber cycle for determination of lattice energy?
(A) Hydration energy of ion
(B) Electron gain enthalpy
(C) Lattice energy
(D) Electronegativity

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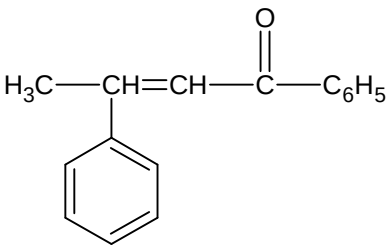
50. The major product in the following reaction is



(A)

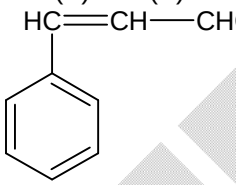


(B)



(C) Both (A) and (B) form with equal percentage

(D)

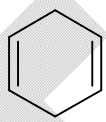


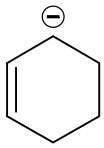
SECTION – B

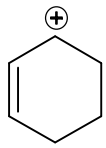
(Numerical Answer Type)

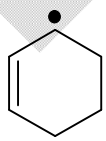
This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

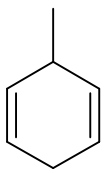
51. How many of the following species can show resonance

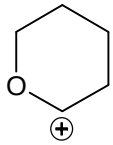
(1) 

(2) 

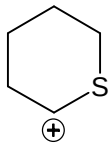
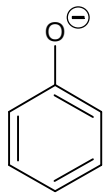
(3) 

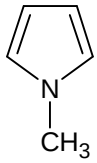
(4) 

(5) 

(6) 

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(7) $CH_2 = CH - Cl$ (8)  (9) 

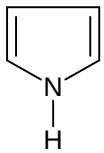
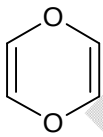
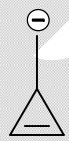
(10) $CH_2 = CH - CH_2 - Cl$ (11) 

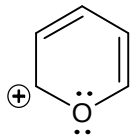
52. HCHO disproportionate to $HCOO^-$ and CH_3OH in presence of OH^-

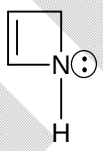
$2HCHO \xrightarrow{OH^-} HCOO^- + CH_3OH$

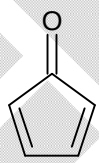
If the equivalent weight of HCHO is E, then the value of E/10 is

53. How many of the following compound is/are aromatic?

(1)  (2)  (3) 

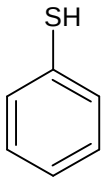
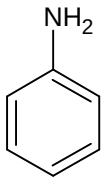
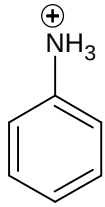
(4)  (5)  (6) 

(7)  (8)  (9) 

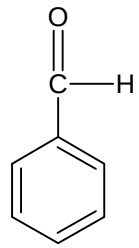
(10) 

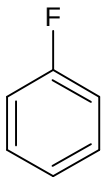
54. How many metal, metal oxide and metal hydroxide are amphoteric in nature
Zn, ZnO, PbO, Pb, PbO_2, FeO, Fe_2O_3, CrO, Cr_2O_3, Fe, Sn, $Zn(OH)_2$

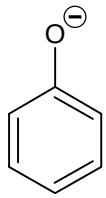
55. How many groups (attached with Benzene ring) can show +M effect?

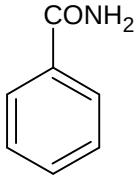
(1)  (2)  (3) 

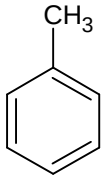
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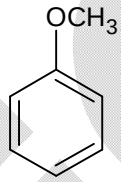
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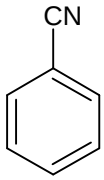
(5) 

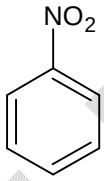
(6) 

(7) 

(8) 

(9) 

(10) 

(11) 

56. How many molecule/ion do not forms either $p\pi - p\pi$ or $p\pi - d\pi$ bond.
SO_3^{2-}, CO_3^{2-}, PO_4^{3-}, SO_3, H_2O, NH_3, $S_2O_3^{2-}$, CH_4, XeF_4

57. Metal (irrespective of oxidation state) present in haemoglobin is also present in
Chlorophyll, Vitamin B_{12}, cis-platin, Sodium Nitroprusside, Brown ring complex, Haematite, Magnetite, Lapis lazuli, Lazurite.

58. In List-I and List-II how many metal ore and process in commercial extraction of metal are correctly matched

List-I **Metal Ore**		**List-II** **Process involved in commercial extraction**
(1) Cinnabar	(P)	Froath floatation method
(2) Chalcopyrite	(Q)	Roasting
(3) Bauxite	(R)	Calcination
(4) Argentite	(S)	Leaching
(5) Haematite	(T)	Carbon reduction method
(6) Coper pyrites	(U)	Self reduction
(7) Carnalite	(V)	Electrolytic reduction
(8) Bauxite	(W)	Electrolytic reduction
(9) Sylvanite	(X)	Cyanide process

59. The sum number of neutrons and protons in radioactive isotopes of hydrogen is

60. Saturated hydrocarbon gas is evolved by hydrolysis of metal carbide(s) among the following is/are
CaC_2, Al_4Cl_3, Mg_2C_3, Be_2C, SiC, Na_2C_2, LaC_2, B_4C, MgC_2

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Mathematics

PART – C

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. The unit vector which is orthogonal to the vector $5\hat{i}+2\hat{i}+4\hat{k}$ and is coplanar with the vector $3\hat{i}-2\hat{i}+\hat{k}$ and $\hat{i}-2\hat{j}+\hat{k}$ is

(A) $\frac{-2}{\sqrt{5}}\hat{j}+\frac{1}{\sqrt{5}}\hat{k}$ (B) $\frac{2}{\sqrt{5}}\hat{j}+\frac{1}{\sqrt{5}}\hat{k}$

(C) $\hat{i}-\frac{2}{\sqrt{5}}\hat{j}+\frac{1}{\sqrt{5}}\hat{k}$ (D) $-\hat{i}+\frac{2}{\sqrt{5}}\hat{j}+\frac{1}{\sqrt{5}}\hat{k}$

62. If the solution of the differential equation $\frac{\sqrt{x}dx+\sqrt{y}dy}{\sqrt{x}dx-\sqrt{y}dy}=\frac{\sqrt{y^3}}{\sqrt{x^3}}$ is $\frac{1}{2}\ln f(x, y) + g(x, y) = c$, then which of the following is true ?

(A) $f(x,y)=\frac{3}{2}(x^3+y^3), g(x,y)=2\sin^{-1}\left(\frac{y}{x}\right)^{3/2}$

(B) $f(x,y)=(x^3+y^3), g(x,y)=2\sin^{-1}\left(\frac{y}{x}\right)^{3/2}$

(C) $f(x,y)=\frac{3}{2}(x^3+y^3), g(x,y)=2\tan^{-1}\left(\frac{y}{x}\right)^{3/2}$

(D) $f(x,y)=(x^3+y^3), g(x,y)=\tan^{-1}\left(\frac{y}{x}\right)^{3/2}$

63. Let $L=\lim_{x\to 0}\frac{(1-\cos 2x)^2(e^{ax}-1)}{x^2(2\tan x-\tan 2x)}=4$, then the value of 'a' given by

(A) 0 (B) 1

(C) –2 (D) 4

64. Let $f(x)=\sum_{k=1}^{n}\text{cosec}2^k x, x\in\left(0,\frac{\pi}{2}\right)$ and $g(x) = f(x) + \cot 2^n x$. If

$$H(x)=\begin{cases}(\cos x)^{g(x)}+(\sec x)^{\text{cosec}x} & , \quad x<0\\ k & , \quad x=0\\ \frac{e^x+e^{-x}-2\cos x}{x\sin x} & , \quad x>0\end{cases}, \text{ then}$$

(A) H is continuous at x = 0 if k = 2
(B) H is continuous at x = 0 if k = 1
(C) H is continuous at x = 0 if k = 0
(D) there is no value of k which can made H continuous at x = 0

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65. $e^{f(x)} = \frac{2+x}{2-x}$, $x \in (-2, 2)$ and $f(x) = k.f\left(\frac{4x}{2+\frac{x^2}{2}}\right)$. then k = ?

(A) 1/4 (B) 1/2
(C) 1 (D) 2

66. The minimum length of chord of circle $x^2 + y^2 - 10x - 10y + 25 = 0$ passing through (2, 2) is

(A) $3\sqrt{7}$ (B) $4\sqrt{7}$
(C) $2\sqrt{7}$ (D) 7

67. Two point (α, α^2) lies on or inside triangle formed by lines $x^2y + xy^2 - 2xy = 0$, then largest value of α

(A) 1 (B) 2
(C) 3/2 (D) 1/2

68. The minimum length of intercept on any tangent to ellipse $\frac{x^2}{4} + \frac{y^2}{9} = 1$ cut by the circle $x^2 + y^2 = 25$

(A) 8 (B) 9
(C) 10 (D) 6

69. A hyperbola having transverse axis of length $2\sin\theta$ is confocal with ellipse $3x^2 + 4y^2 = 12$ then its equation is

(A) $x^2\sin^2\theta - y^2\cos^2\theta = 1$ (B) $x^2\tan^2\theta - y^2\cot^2\theta = 1$
(C) $x^2\sec^2\theta - y^2\operatorname{cosec}^2\theta = 1$ (D) $x^2\operatorname{cosec}^2\theta - y^2\sec^2\theta = 1$

70. Let P(x) be a polynomial of degree 4, such that P(–1) = 0 and has local maximum or local minimum at x = 1, 2, 3 also $\int_{-2}^{2} P(x)dx = \frac{-1348}{15}$, then the y-intercept of the tangent to curve y = p(x) at x = –1 equals to

(A) 24 (B) –8
(C) –96 (D) 32

71. Six identical balls are randomly dropped into three boxes A, B, C which can hold any number of balls. If the probability that one of the box contain 1 and other two contain 2 and 3 balls is

(A) $\frac{5}{14}$ (B) $\frac{9}{14}$
(C) $\frac{3}{14}$ (D) $\frac{2}{7}$

72. If given equation is (3 + sint) (3 + cost) = 10 where $t \in R$, then the value of sin2t is

(A) $4\left(5+3\sqrt{3}\right)$ (B) $4\left(5-3\sqrt{3}\right)$
(C) $2\left(5-3\sqrt{3}\right)$ (D) $\left(5-3\sqrt{3}\right)$

73. The minimum value of the function $f(x) = \frac{2\sin x}{\sqrt{1-\cos^2 x}} + \frac{4\cos x}{\sqrt{1-\sin^2 x}} + \frac{8\tan x}{\sqrt{\sec^2 x - 1}} + \frac{16\cot x}{\sqrt{\operatorname{cosec}^2 x - 1}}$ whenever it is defined is

(A) –34 (B) –30
(C) –26 (D) –22

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74. $A=\begin{bmatrix}3 & 5 & -7\\ -2 & 10 & 5\\ 9 & 6 & 14\end{bmatrix}, B=\begin{bmatrix}1 & 1 & -1\\ 2 & 2 & -3\\ -1 & 0 & 2\end{bmatrix}$ and $\begin{bmatrix}4 & -2 & -1\\ -1 & 1 & 1\\ 2 & -1 & 0\end{bmatrix}$, then value of infinite series

$S=\text{Tr}(A)+\text{Tr}\left(\frac{3}{4}A(BC)\right)+\text{Tr}\left(\frac{6}{4^2}A(BC)^2+\text{Tr}\left(\frac{10}{4^3}A(BC)^3\right)\right)+.....\infty$ is

(A) 64 (B) 27
(C) 28 (D) 32

75. Let x_i be 15 observations where $1 \le i \le 15$. If $\sum_{i=1}^{15}(x_i-\lambda)=8$ and $\sum_{i=1}^{15}(x_i-\lambda)^2=12$, then standard deviation of these observation is

(A) $\frac{2\sqrt{28}}{15}$ (B) $\frac{116}{225}$
(C) $\frac{2\sqrt{29}}{15}$ (D) $\frac{112}{225}$

76. Let f be a continuous function on [0, 1] and differentiable on (0, 1) with $f'(x) = e^{x^2}$ and f(0) = 4, then

(A) $2 < f(1) < 3$ (B) $3 < f(1) < 4$
(C) $5 < f(1) < 4 + e$ (D) $2 < f(1) < 4$

77. A twice differentiable function $f : R \to R$ satisfies
$\sin x\cos y(f(2x + 2y) - f(2x - 2y)) = \cos x\sin y(f(2x + 2y) + f(2x - 2y))$
for every $x, y \in R$, then which of following is true ?

(A) $f''(x) - f(x) = 0$ (B) $4f''(x) + f(x) = 0$
(C) $f''(x) + f(x) = 0$ (D) $4f''(x) - f(x) = 0$

78. The value of θ such that $0 < \theta < \frac{\pi}{2}$ and satisfying the equation

$$\begin{vmatrix}1+\sin^2\theta & \cos^2\theta & 4\sin 4\theta\\ \sin^2\theta & 1+\cos^2\theta & 4\sin 4\theta\\ \sin^2\theta & \cos^2\theta & 1+4\sin 4\theta\end{vmatrix}=0 \text{ is}$$

(A) $\frac{5\pi}{24}$ (B) $\frac{\pi}{24}$
(C) $\frac{\pi}{8}$ (D) $\frac{7\pi}{24}$

79. The maximum value of the expression $\frac{1}{2}\sum_{n=3}^{20}{}^nC_3\left(\sin^{2n}x+\text{cosec}^{2n}x\right)$ is

(A) ${}^{21}C_4$ (B) ${}^{20}C_3$
(C) ${}^{21}C_5$ (D) 2^{20}

80. If the distance between the plane $Ax + y - z = d$ and the plane containing the line $\frac{x}{1}=\frac{y-2}{-1}=\frac{z-1}{1}$ and $\frac{x-2}{2}=\frac{y-3}{1}=\frac{z-6}{5}$ is $\sqrt{6}$, then $|d - 1| + A$ is

(A) 7 (B) 6
(C) 5 (D) 8

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SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

81. If $I=\int_0^{\infty}\frac{2x^2}{e^x-e^{-x}}dx$ and $J=\int_0^{\infty}\frac{x^2}{e^x-1}$ then $\frac{4I}{j}$

82. If t_1, t_2, t_3, t_4 and t_5 are first 5 terms of an A.P., then $\frac{4(t_1-t_2-t_4)+6t_3+t_5}{3t_1}$ is equal to

83. Area of region formed by point (x, y) satisfying $[x]^2 = [y]^2$ for $0 \le x \le 4$ is equal to

84. If the equation of parabola $a(x - y + 1)^2 = b(x + y - 4)$ where a and b are relatively prime natural numbers and $y = 2x + 3$ is one of its tangent then $a + b =$

85. Let $f(x)=\sqrt{\sin^4 x+4\cos^2 x}-\sqrt{\cos^4+4\sin^2 x}$ and $g(\sin 2t) = \sin t + \cos t$, $\forall\ t \in \left[-\frac{\pi}{4},\frac{\pi}{4}\right]$. If the range of g(f(x)) is [a, b], then the value of $a^2 + b^2$ is

86. Let x, y, z be positive numbers such that $x + y + z = 10$. If maximum value of $x^2y^3z^5$ is N, then number of divisors of N which are of the form $(4n + 2)$, $n \in N$ is

87. If $\vec{a},\vec{b},\vec{c}$ and $\vec{d}$ are the position vectors of the point A, B and C and D respectively, in three dimensional space and satisfy the relation $3\vec{a}-2\vec{b}+\vec{c}-2\vec{d}=0$, the line joining the points A and C divides the line joining the points B and D in the ratio x : y, then vale of (x + y)

88. Number of common solutions of the equations $2\sin^2\theta - \cos 2\theta = 0$ and $\sqrt{3}\sin 2\theta = 2\cos^2\theta$ in $\theta \in \left[-\frac{\pi}{2},\frac{5\pi}{2}\right]$.

89. Roots of the equation $1 + {}^nC_1z + {}^nC_2z^2 + {}^nC_3z^3 + \ldots\ldots + {}^nC_{n-1}z^{n-1} = 0$ lie on curve 'C' in Argand plane then the least distance between curve 'C' and curve $\left|z-\frac{5}{2}\right|=1$ is

90. If a function $f(x) = ax^3 + bx^2 + cx + d$, where a, b, c and d are integers and $a > 0$, is such that $f\left(\sin\frac{\pi}{18}\right)=0$, then smallest possible value of f(1) is

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